

- 4 (a) Three capacitors are connected as shown in Fig. 4.1.

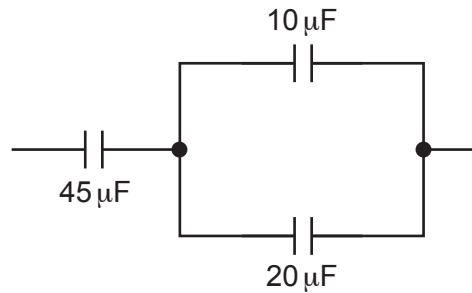


Fig. 4.1

Determine the total capacitance, in μF , of the network of three capacitors.

capacitance = μF [2]

- (b) A capacitor of capacitance $45\ \mu\text{F}$ is connected to a variable power supply initially set at $8.0\ \text{V}$. The output of the power supply increases so that the potential difference (p.d.) across the capacitor increases to $9.6\ \text{V}$.

Calculate the increase in energy ΔE stored in the capacitor.

$\Delta E = \dots\dots\dots\ \text{J}$ [2]

- (c) A sinusoidal a.c. power supply is connected to the input of a bridge rectifier. The output of the rectifier is connected to a load resistor.
- (i) Complete the circuit in Fig. 4.2 by adding a capacitor to smooth the p.d. across the load resistor.

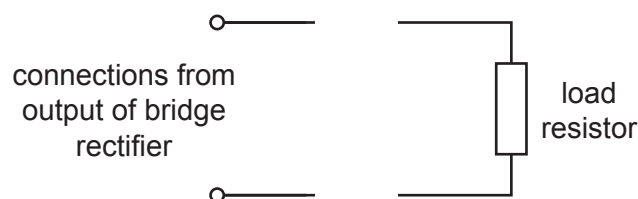


Fig. 4.2

[1]

(ii) The variation with time t of the p.d. V of the smoothed output is shown in Fig. 4.3.

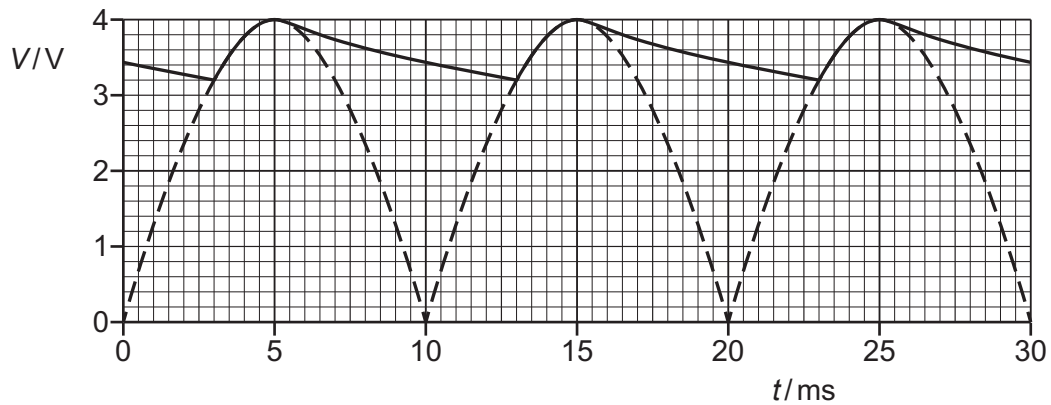


Fig. 4.3

Determine the time constant, in ms, of the smoothing circuit.

time constant = ms [3]

(d) A sinusoidal a.c. power supply has a maximum power of 16 W.

State the value of the mean power when the output of the power supply is:

(i) full-wave rectified

mean power = W [1]

(ii) half-wave rectified.

mean power = W [1]

[Total: 10]